CENTINELA

IEEE MYOSA Event 6.0 Proposal

A. Statement of Interest

Our team is pleased to submit CENTINELA, a wearable safety bracelet designed for agricultural workers operating alone in large farms. The system uses the MYOSA platform as its computational core. An ESP32 executes a TinyML model that analyzes inertial data from the MPU6050 to detect abnormal immobility. Before transmitting an emergency, the worker has a confirmation window that can cancel false alarms through a gesture detected by the APDS-9960. Once confirmed, an alert is sent through a LoRa link to a base station and displayed in a web-based control tower. The prototype demonstrates edge AI, embedded systems, low-power operation and real-time sensing while addressing a real occupational safety problem.

B. Team Members

Role	Name	Details	Year
Leader	José David Quiñonez Rehenals	Biomedical Engineering, Universidad Tecnológica de Bolívar	3
Member	Junior José Sarmiento Prada	Biomedical and Mechatronics Engineering, Universidad Tecnológica de Bolívar	4

Leader email: j.quinonez0810@gmail.com
Leader phone: +57 3127368737
Member email: juniorjosesarmientoprada@gmail.com
Member phone: +57 3004010653

C. Faculty Mentor

Name: Alberto Patiño Vanegas
Institution: Universidad Tecnológica de Bolívar
Email: apatino@utb.edu.co
Phone: +57 318 8315779

Application Significance

Agricultural workers are frequently exposed to heat stress, dehydration and falls while working without direct supervision. CENTINELA provides autonomous local monitoring using embedded artificial intelligence and only transmits alerts when necessary, reducing infrastructure requirements and improving response time.

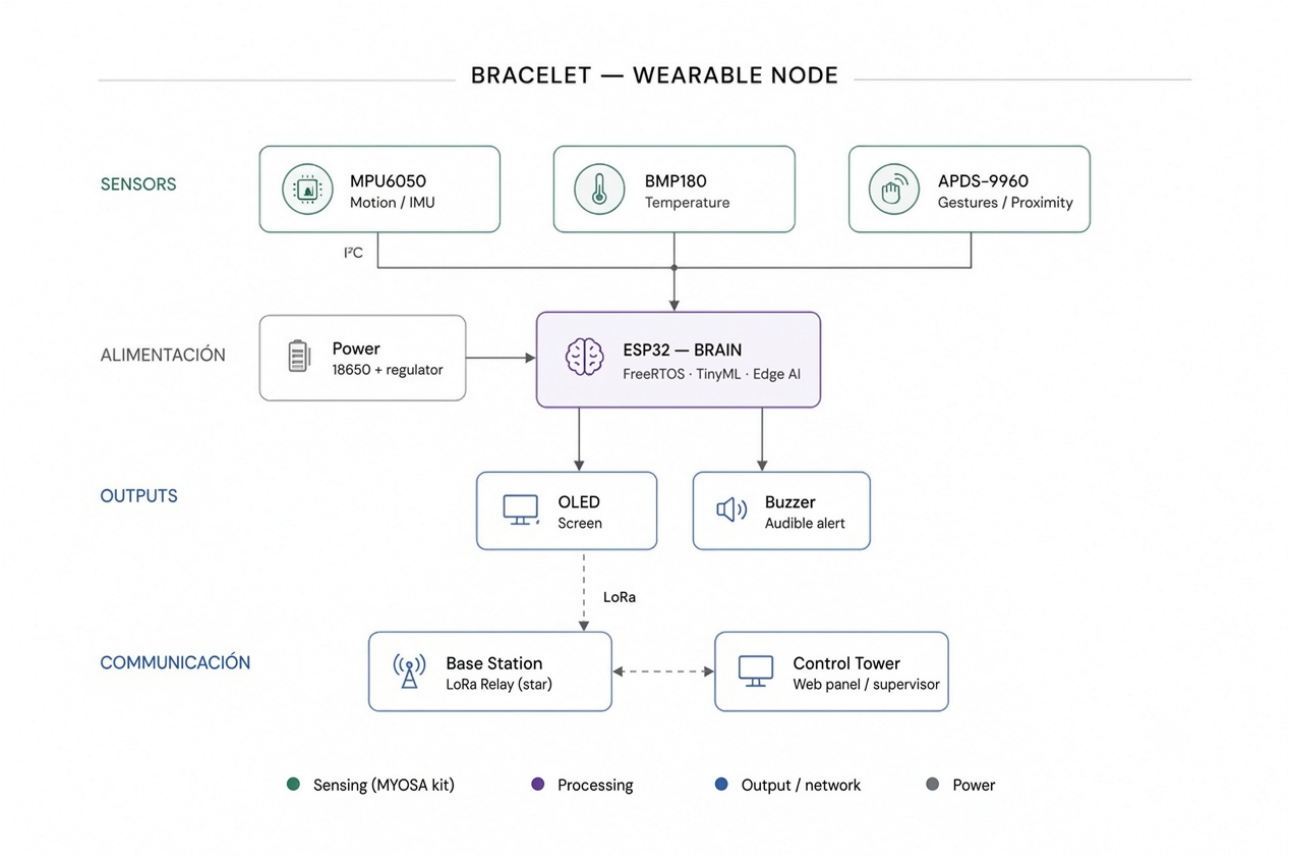
MYOSA Platform

The ESP32 performs signal acquisition, preprocessing and TinyML inference. The OLED provides local feedback while the APDS-9960 allows gesture-based interaction. A LoRa module extends communication to a remote base station.

System Components

Component	Function
MPU6050	Motion sensing for TinyML classifier.
BMP180	Ambient temperature context.
APDS-9960	Gestures, wake-up and alarm cancellation.
OLED	Local diagnostics.
Buzzer	Audible emergency warning.
LoRa	Long-range communication.
18650 Battery	Portable power supply.

System Architecture



Demonstration

The live demonstration will show normal operation, abnormal immobility detection, user confirmation, alarm transmission through LoRa and visualization in the control dashboard.

Validation and Limitations

The classifier will be evaluated using controlled activity scenarios. The prototype estimates worker status from wrist motion only, uses area-based localization instead of GPS and demonstrates proof-of-concept communication.